

Polynomial Curve Fitting: A Comparison of Three Linear System Cases

Special Recognition To:
GEARS Research Program
Department of Mathematics



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The Problem

- A continuous function is often estimated from finitely many sampled values.
- This project approximates $f(t) = \sin(at)$ on $0 \leq t \leq \pi/2$ with a polynomial.
- The values $m + 1$ and n create over-, exactly-, or under-determined systems.
- We study accuracy and instability as the sample count, degree, and a change.

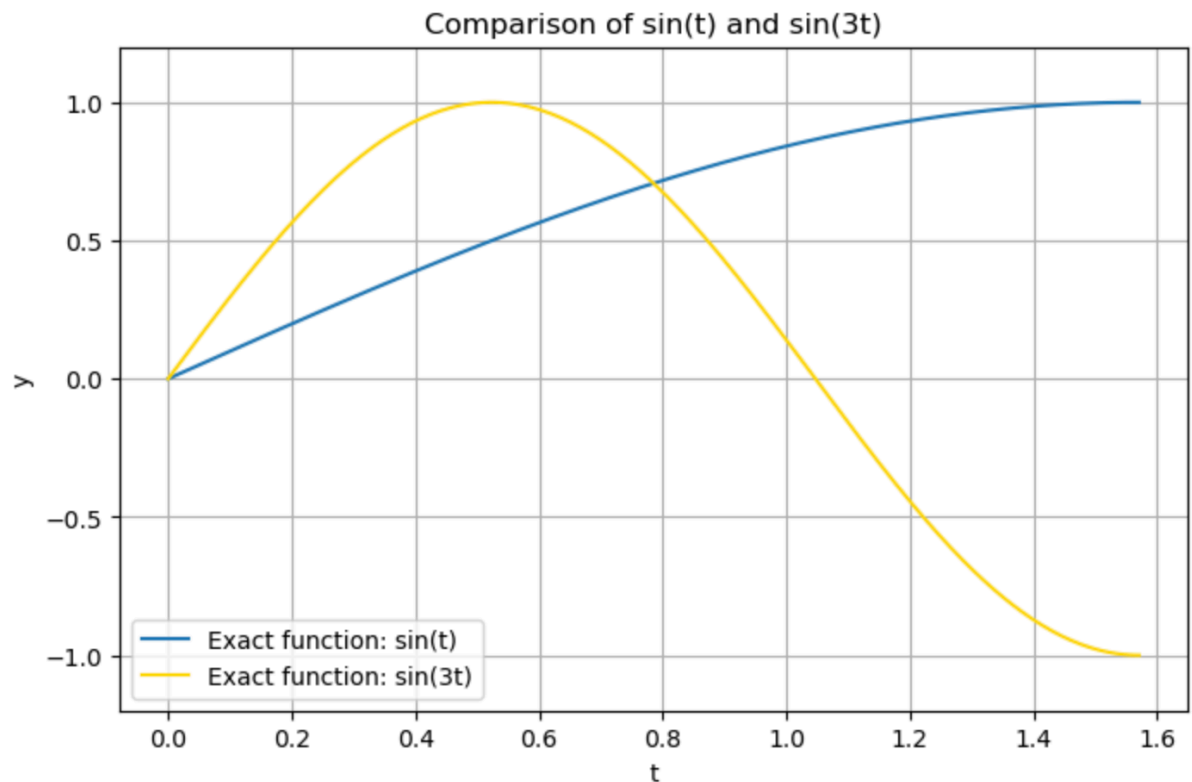
How number of sample point ($m+1$), polynomial degree(n), frequency parameter(a) affect accuracy & numerical stability.

Polynomial Approximation

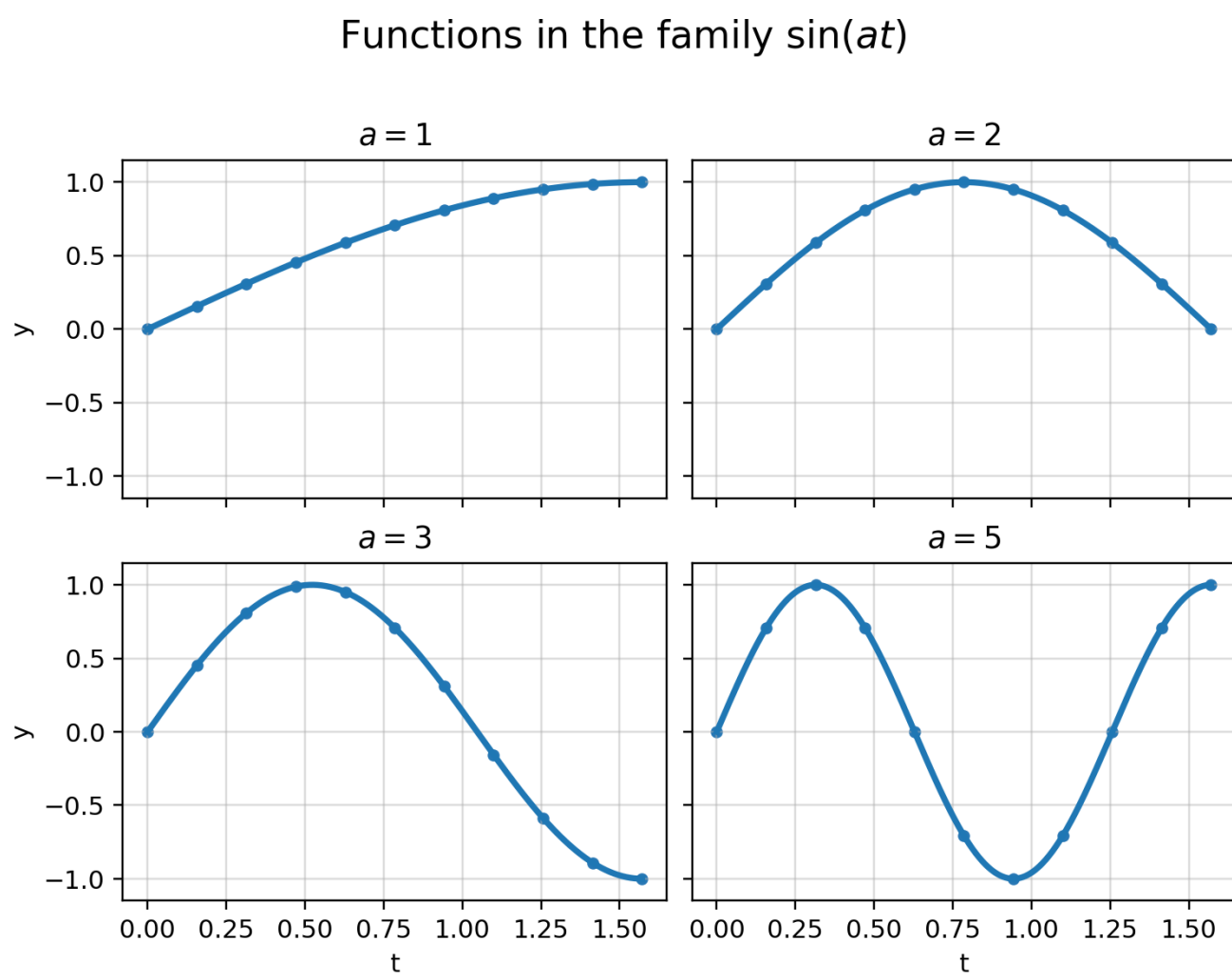
- The parameter a changes frequency while the amplitude remains between -1 and 1 .

$$f(t) = \sin(at), \quad 0 \leq t \leq \frac{\pi}{2}$$

- The target function is sampled at equally spaced points before the coefficients are computed.



- With enough data and a moderate degree, the fitted curve can closely follow the true function.



$a \uparrow \Rightarrow$ faster oscillation $\Rightarrow$ larger n is usually needed

- A family of functions $\sin(at)$ was tested for several values of a .
- Increasing a produces faster changes on the same interval.
- This motivates comparing accuracy, model degree, and numerical stability.

Therefore, the larger 'a' is, the more peaks, troughs, turns, and intersections with the horizontal axis the curve needs to be fitted. The polynomial degree limits the number of turns.

Mathematical Formulation

- Sample the target function at $m + 1$ equally spaced points.

$$t_i = \frac{i\pi}{2m}, \quad y_i = \sin(at_i), \quad i = 0, \dots, m$$

- Represent the approximation by a polynomial of degree n .

$$p_n(t) = c_0 + c_1 t + \dots + c_n t^n = \sum_{j=0}^n c_j t^j$$

$$A_{ij} = t_i^j, \quad A\mathbf{c} \approx \mathbf{y}$$

- The relation between m and n selects the solution method.

$m > n$: least squares $m = n$: interpolation $m < n$: minimum-norm solution

$$m > n: \quad \min_{\mathbf{c}} \|A\mathbf{c} - \mathbf{y}\|_2$$

$$m = n: \quad A\mathbf{c} = \mathbf{y}, \quad \mathbf{c} = A^{-1}\mathbf{y}$$

$$m < n: \quad \mathbf{c} = A^+ \mathbf{y}$$

- Evaluate the fitted polynomial on a dense grid and measure its error.

$$\hat{y}(t) = p_n(t) = (1 \ t \ \dots \ t^n) \mathbf{c}$$

$$E_{\max} = \max_t |p_n(t) - \sin(at)|$$

Prediction quality depends on m, n, a , and conditioning of A .

Examples/Results

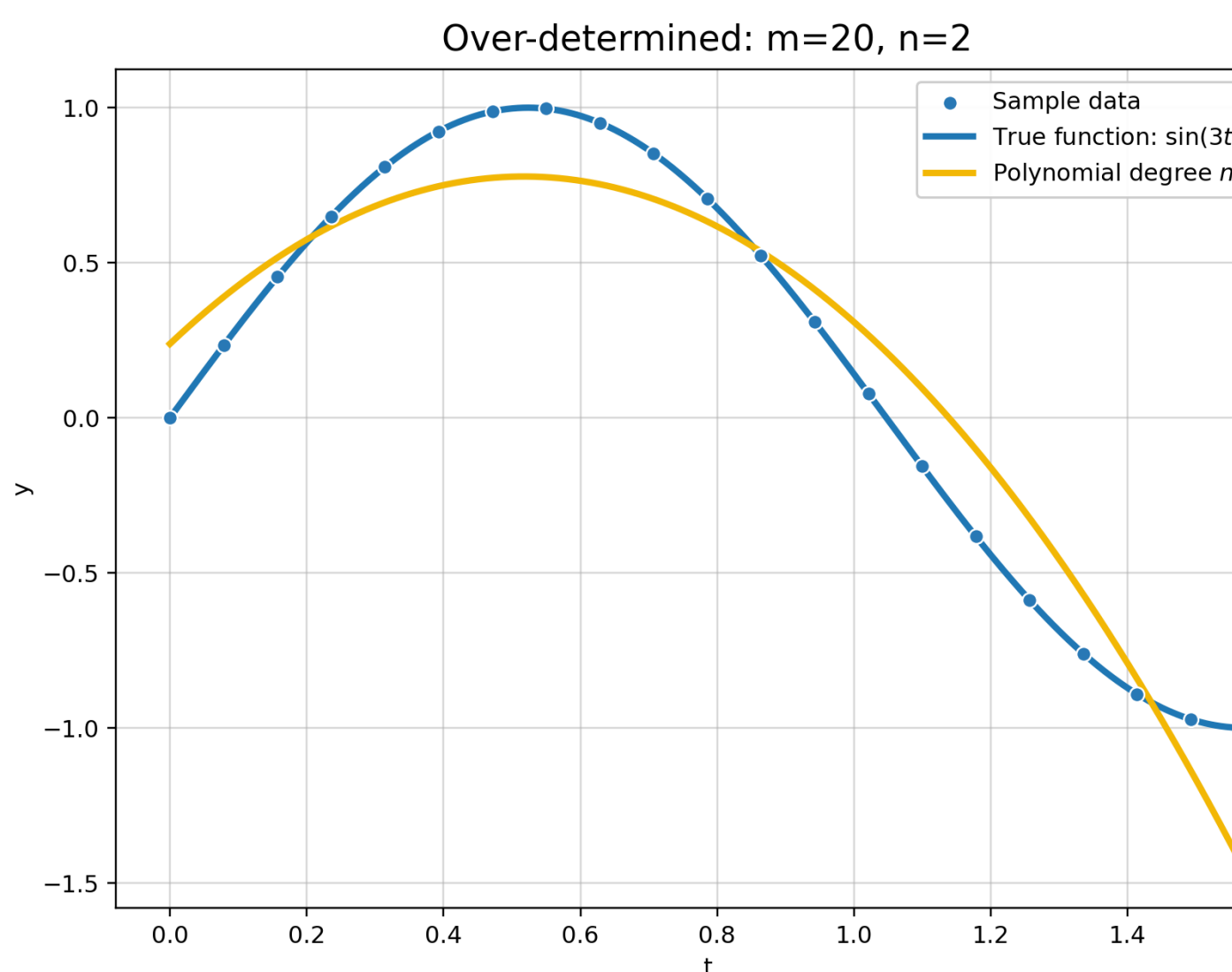


Figure 1: Degree 2 underfits $\sin(3t)$, although all 21 samples are used.

$$m = 20, n = 2: \quad E_{\max} \approx 0.459$$

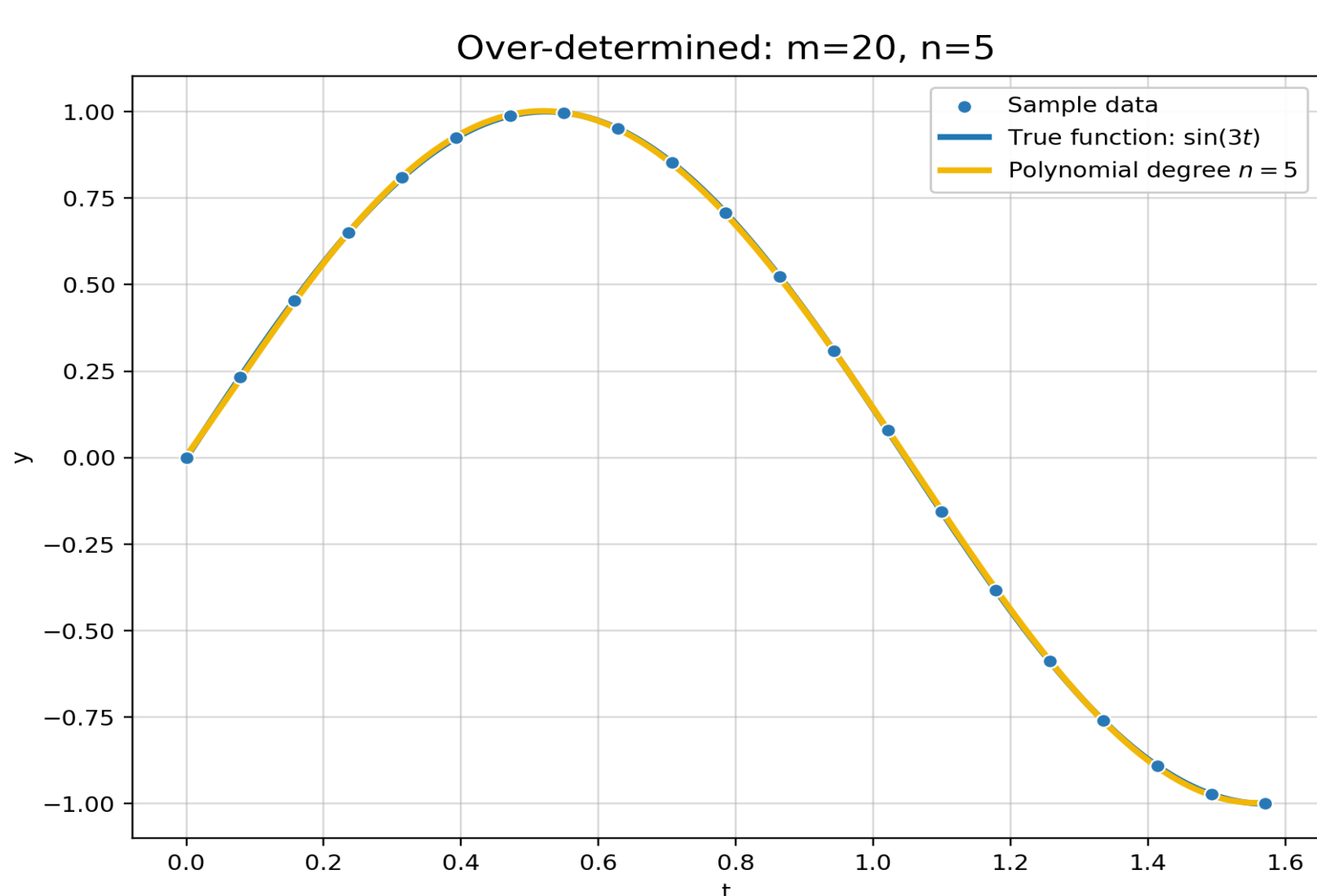


Figure 2: Degree 5 gives a stable, accurate least-squares approximation.

$$m = 20, n = 5: \quad E_{\max} \approx 0.00585$$

$n \uparrow$: approximation improves, but $\kappa(A)$ also grows.

References

Li, Z. (2026). *Polynomial approximation and least-squares fitting [Course notes]. GEARS Program, North Carolina State University.*

Popiński, W. (2020). *Least squares orthogonal polynomial regression estimation for irregular design. Communications in Statistics—Theory and Methods*, 49(3), 631– 647. <https://doi.org/10.1080/03610926.2018.1549244>

Examples/Results

- Exactly determined interpolation and numerical conditioning

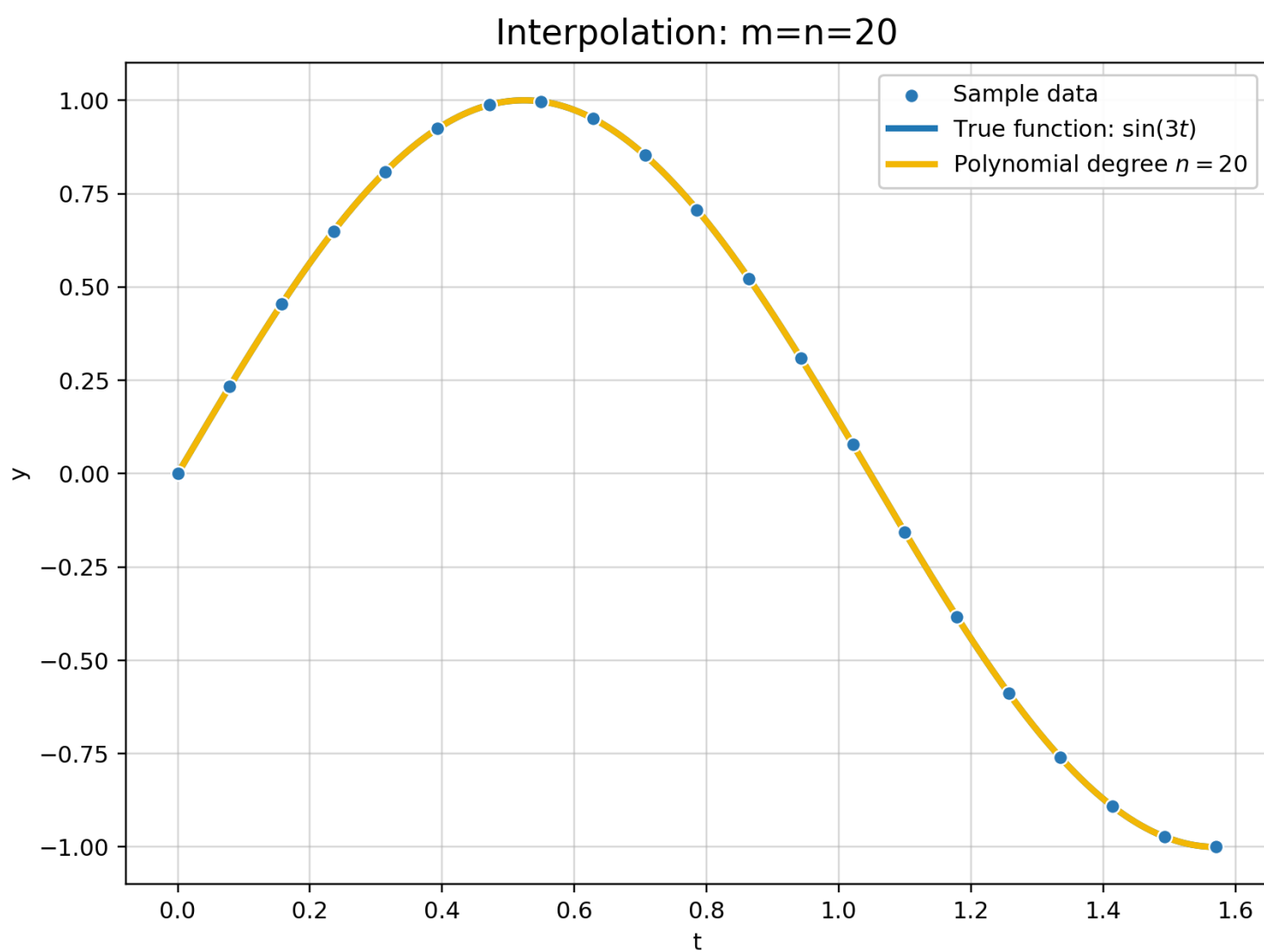
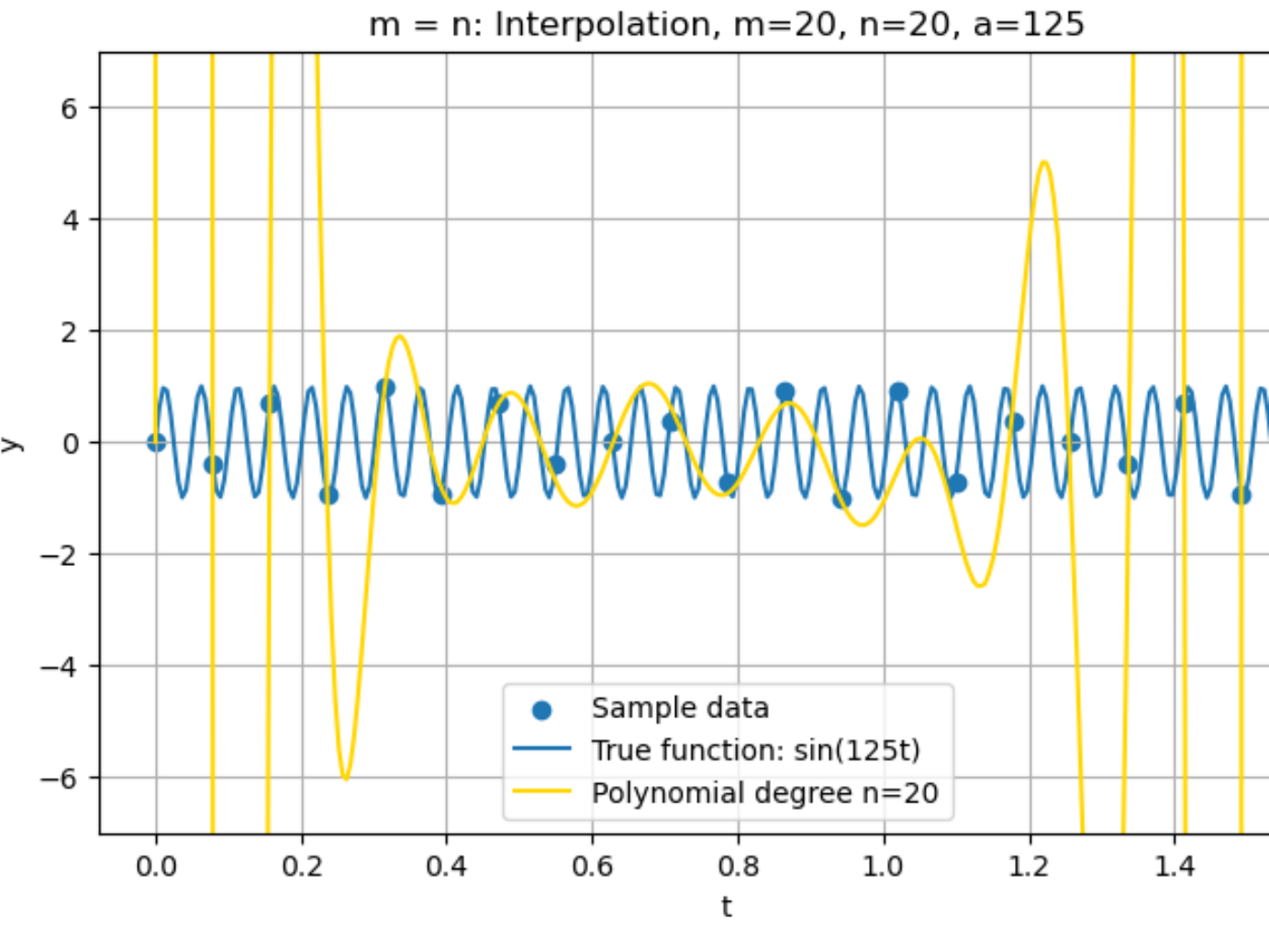


Figure 3: The degree-20 polynomial passes through all 21 samples.



$$E_{\max} \approx 1.24e - 12$$

$$m = n = 20$$

$$\kappa(A) \approx 4.34e + 17$$

- Under-determined high-degree approximation

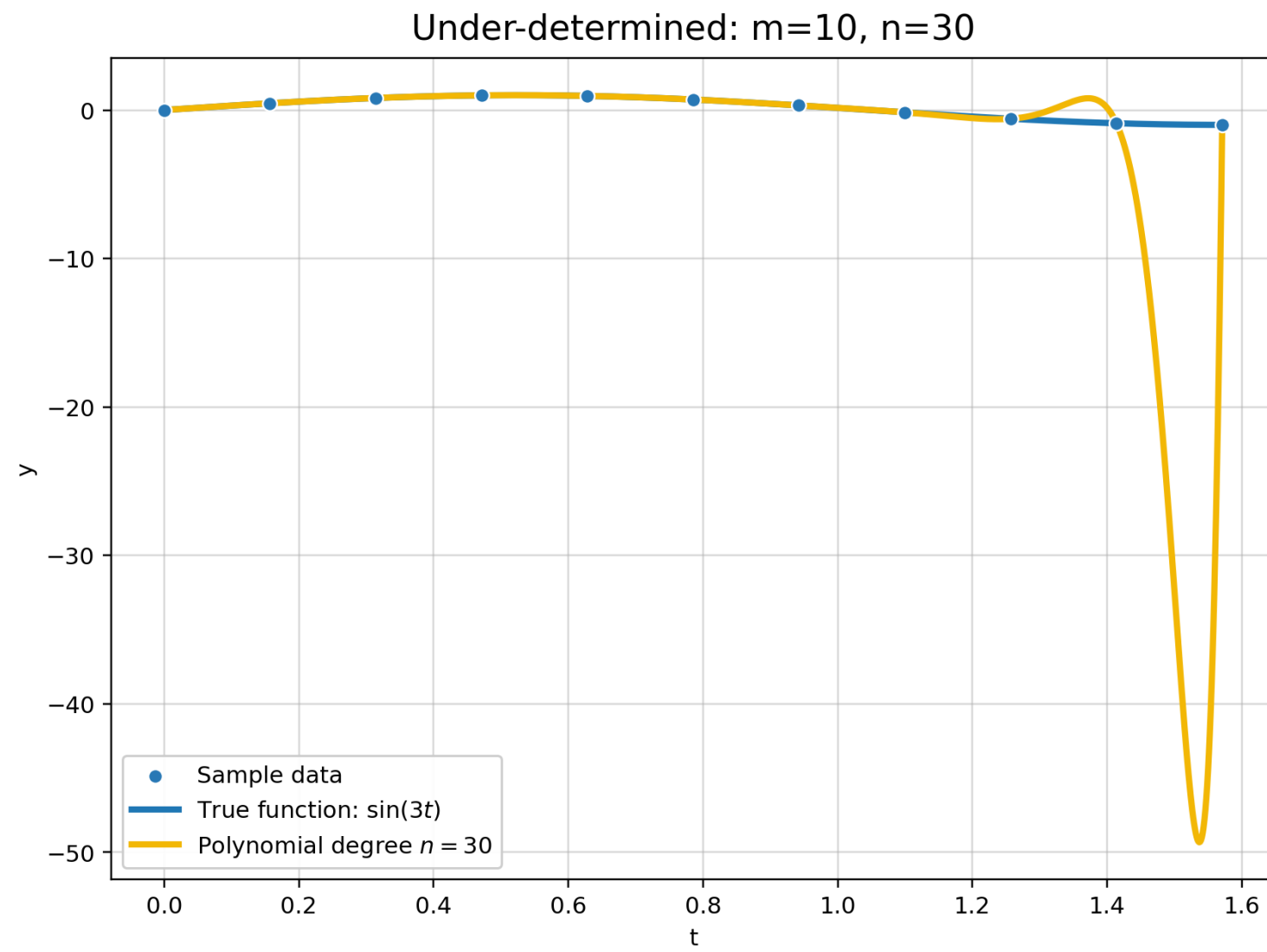


Figure 5: A degree-30 minimum-norm fit becomes unstable between samples.

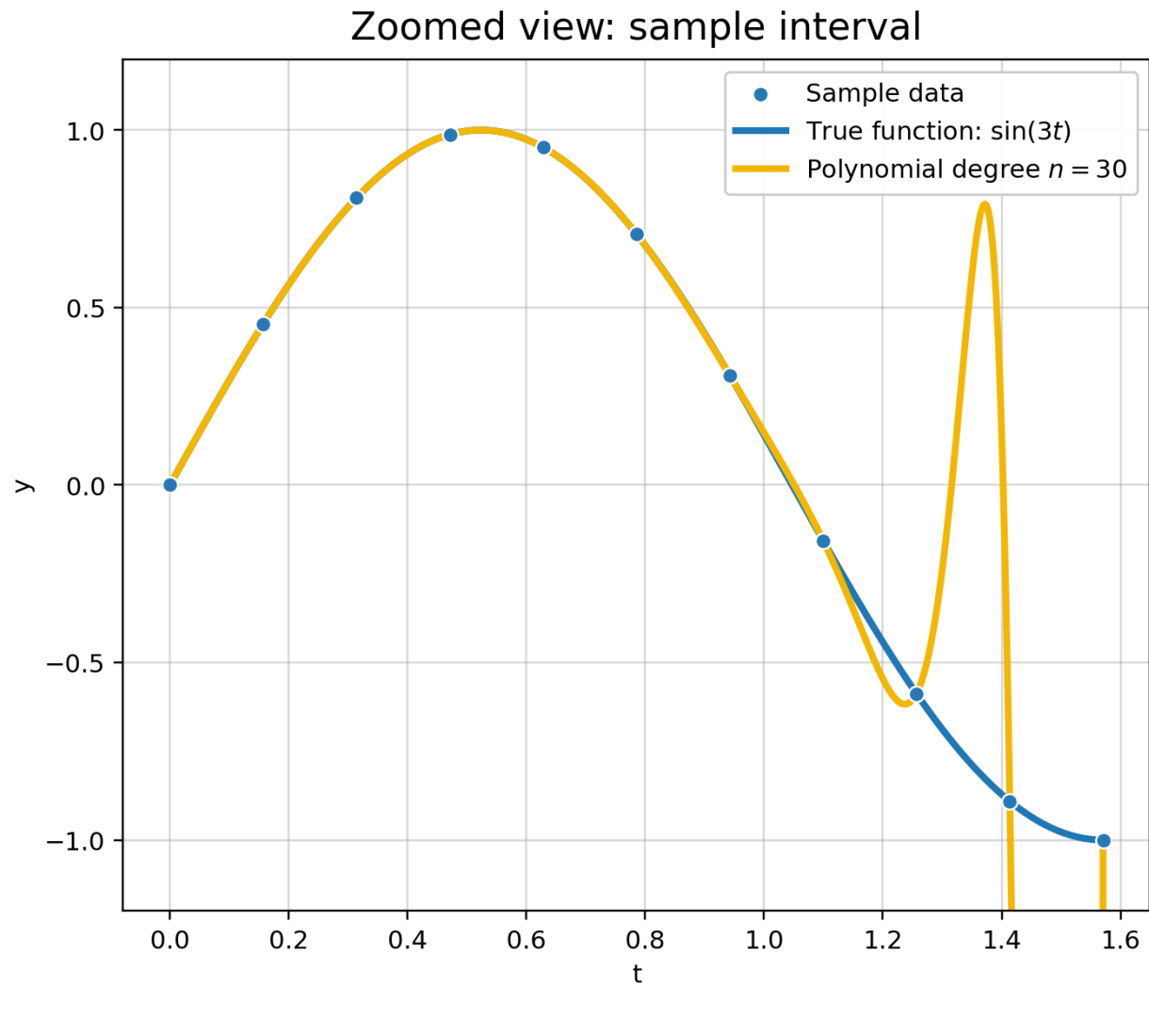


Figure 6: The zoomed view confirms the true function remains in $[-1, 1]$.

$$\max_i |p_n(t_i) - y_i| \approx 2.96e - 09$$

$$m = 10, n = 30$$

$E_{\max} \approx 48.3$: small sample residual does not guarantee stable prediction.

Conclusion and future work

Increasing the polynomial degree can improve accuracy, but it may also increase the condition number and cause numerical instability. A polynomial may fit all sample points while remaining inaccurate between them. Therefore, polynomial approximation requires a balance between degree, sampling density, and stability.

Future Work

Future work will extend the experiments to cosine functions. The final stage will use pandas to process real historical stock-price data and test polynomial fitting on real-world time-series data.